can click on one of the traces and land in a view shown in figure 6.7. The trace details view gives you the individual spans a trace is composed of in the waterfall model, including duration and high-level metadata, such as type of operation and whether it was successful or not. Let's move on to the another CNCF project in this category.

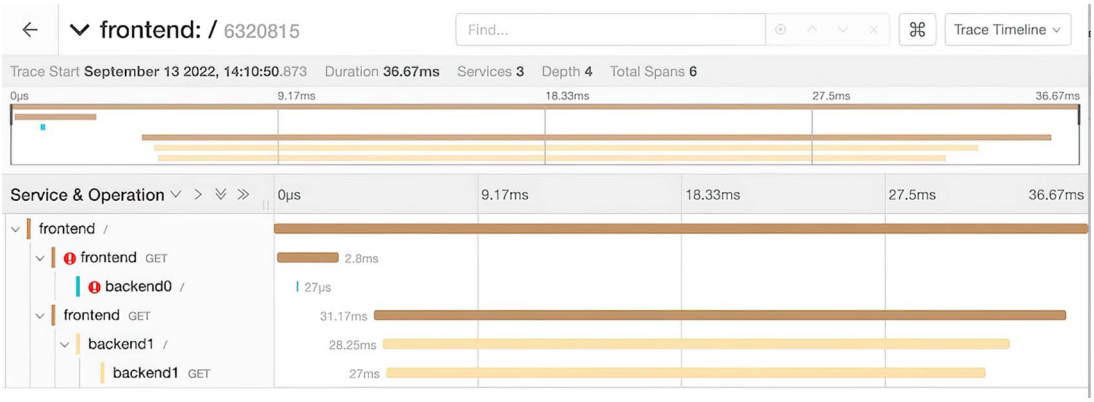


Figure 6.7 Jaeger UI showing trace details

6.2.2 CNCF Pixie

Pixie (https://px.dev/) is a Kubernetes-specific, real-time debugging platform that collects telemetry data, such as metrics, traces, and profiles, without manual instrumentation or redeploys. Pixie has a frontend that can help you query and visualize this data. It was contributed to CNCF by New Relic and accepted as a sandbox project in June 2021.

To collect data without any code changes, Pixie heavily uses eBPF (https://ebpf.io/). eBPF is a Linux kernel feature that allows you to execute sandboxed programs, after verification, directly in the kernel.

Since eBPF allows you to collect large amounts of fine-grained data, such as full request and response bodies, it stores all its data in memory. You can, however, send the telemetry data to any OpenTelemetry-compatible endpoint (http://mng.bz/JgXV) for long-term data retention. In other words, historical data is, by design, out of scope for Pixie.

To query the telemetry data, Pixie uses a rather clever domain-specific language called PxL (https://docs.px.dev/tutorials/pxl-scripts/write-pxl-scripts) that was inspired by Python Pandas. Users can use PxL to query and process Pixie's underlying data however they like. All of Pixie's dashboards are also entirely scriptable, allowing users to build the views and visualizations that work best for them.

Pixie has a lively community with an interesting and promising road map. It sports a number of integrations, from Slack alerts to using it in a Grafana dashboard (via OpenTelemetry export). If you're running a Kubernetes cluster, I recommend taking